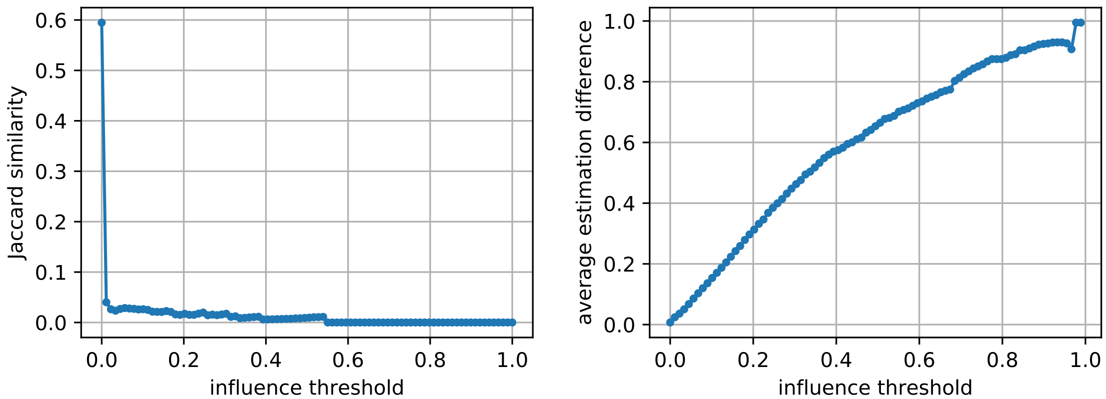

Figure 8: Consistency of the estimation of influence between ResNet50 and linear models trained on the penultimate layer representations computed on the entire CIFAR-100 dataset.

same class. Despite the lack of memorization, we still found 457 high-influence pairs of examples (as before, those with influence estimates above 0.15). In most of these pairs we see no visual similarity (although there is still a significant fraction of those that are visually similar). In Fig. 8 we quantify the large discrepancy in estimates obtained using this technique and training of the full ResNet50 model. Specifically, we compare the influence estimates using the Jaccard similarity and average difference in estimate but without the additional constraint of having memorization value above 0.25 (since it is satisfied only by 38 examples).

4 Discussion

Our experiments provide the first empirical investigation of memorization and its effects on accuracy that are based on formally defined and intuitive criteria. The results reveal that, in addition to outliers and mislabeled examples, neural networks memorize training examples that significantly improve accuracy on visually similar test examples. These pairs of examples are visually atypical and most train and test examples only appear in a single pair. This, we believe, strongly supports the long tail theory and, together with the results in [Fel19], provides the first rigorous and compelling explanation for the propensity of deep learning algorithms to memorize seemingly useless labels: it is a result of (implicit) tuning of the algorithms for the highest accuracy on long-tailed and mostly noiseless data.

Our work demonstrates that accuracy of a learning algorithm on long tailed data distributions depends on its ability to memorize the labels. As can be seen from the results, the effect on the accuracy of not memorizing examples depends on the number of available examples and the data variability (a formal analysis of this dependence can be found in [Fel19]). This means that the effect on accuracy will be higher on an under-represented subpopulation. The immediate implication is that techniques that limit the ability of a learning system to memorize will have a disproportionate effect on under-represented subpopulations. Techniques aimed at optimizing the model size (e.g. model compression) or training time are likely to affect the ability of the learning algorithm to memorize data. This effect is already known in the context of differential privacy (which formally limits the ability of an algorithm to memorize data) [BPS19].

The experiments in Section 3.6 demonstrate that most of memorization happens in the representations derived by training a DNN. A natural direction for future work is to derive a detailed understanding of the process of memorization by a training algorithm.

The primary technical contribution of our work is the development of influence and memorization estimators that are simple to implement, computationally feasible, and essentially as accurate as true leave-one-out influences. While several other approaches for influence estimation exist, we believe that our approach provides substantially easier to interpret results. Unlike some of the existing techniques [KL17; YKYR18; PLSK20] it is also completely model-agnostic and is itself easy to explain. In addition to understanding of deep learning, influence estimation is useful for interpretability and outlier detection and, we hope, our estimator will find applications in these areas.

Computing our estimator with high accuracy relies on training thousands of models and thus requires significant computational resources. A natural direction for future work is finding proxies for our estimator that can be computed more efficiently. To simplify future research in this direction and other applications of our estimator, we provide the computed values for CIFAR-100 and ImageNet at [FZ20].

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